

BRRRR Analysis Results: Formulas

Transcribed from BackEnd/BL/analyze/brrrSteps/.py and BackEnd/BL/analyze/common/deal_math.py*

Cash Flow

cash_flow.py, operating_expenses.py, mortgage_payment.py

```
cash_flow = payload.rent - operating_expenses - mortgage_payment

operating_expenses =
    payload.rent * payload.vacancy_percent / 100
+ payload.rent * payload.property_management_fee_percentages_from_rent / 100
+ payload.rent * payload.maintenance_percent / 100
+ payload.rent * payload.capex_percent_of_rent / 100
+ payload.annual_property_taxes / 12
+ payload.annual_insurance / 12
+ payload.monthly_hoa

mortgage_payment:
    loan_amount = arv * ltv                      (arv = payload.arv_in_thousands * 1000, ltv = payload.ltv_as_percent / 100)
    monthly_rate = payload.interest_rate / 100 / 12
    n = payload.loan_term_years * 12
    factor = (1 + monthly_rate) ** n
    mortgage_payment = loan_amount * monthly_rate * factor / (factor - 1)
                    = loan_amount / n              if monthly_rate == 0
```

Cash Out

cash_out.py

```
cash_out = cash_out_routi - total_cash_invested

total_cash_invested =
    payload.earnest_money_deposit
+ cash_to_close_buy
+ max(0, seller_tax_credit)                      (seller_tax_credit_set_aside_in_tax_bucket)
+ rehab_paid_cash_out_of_pocket                  (= rehab_cost - construction_budget)
+ hml_and_holding.hml_interest_paid_monthly
+ hml_and_holding.holding_costs
+ hml_and_holding.utilities_until_rented
+ payload.maintenance_before_refi
+ payload.appliances
- hml_and_holding.pre_refi_rental_income
```

Cash-Out Routi

cash_out.py, refi_terms.py

```
cash_out_routi =
    refi_loan_amount                (= arv * ltv)
- hml_payoff                        (= hml_amount + hml_interest_accrued_into_refi_payoff)
- closing_costs_refi_total (at arv)
- prepaid_interest_refi (at arv)
- reserves_total

closing_costs_refi_total (at loan L) =
    payload.loan_charges_refi
+ (payload.recording_transfer_refi or 0.0055 * L + 250)
+ (payload.title_escrow_refi or 800 + 0.0045 * L)
+ (payload.online_notary_fee_refi or 250, if payload.online_notary_refi, else 0)
+ payload.appraisal_fee
+ payload.survey_fee
+ payload.refi_underwriting_fee
+ payload.refi_points / 100 * L
+ payload.broker_processing_fee_refi
+ payload.other_closing_costs_refi

prepaid_interest_refi (at loan L) = L * payload.interest_rate * dscr_interest_days_prepaid_at_refi_closing / 365 / 100
reserves_total = payload.maintenance_reserve + (payload.vacancy_reserve or payload.rent) + payload.capex_reserve
```

CoC

cash_on_cash.py

```
cash_on_cash = -1                if cash_out >= 0
                = -2                if cash_flow <= 0
                = cash_flow * 12 / abs(cash_out) * 100    otherwise
```

DSCR

dscr.py

```
pitia = mortgage_payment + payload.annual_property_taxes / 12 + payload.annual_insurance / 12 + payload.monthly_hoa
dscr = payload.rent / pitia          (0 if pitia == 0)
```

Equity

equity_and_net_profit.py

```
equity = arv * (1 - ltv) + reserves_total
```

ROI

roi.py

```
roi = -1                if cash_out >= 0
    = -2                if cash_flow <= 0
    = (cash_flow * 12 + net_profit) / abs(cash_out) * 100    otherwise
```

Net Profit

equity_and_net_profit.py

```
net_profit = equity + cash_out
```

Cash Needed

total_cash_needed.py

```
cash_needed_through_refi =
    total_cash_invested                (expanded under Cash Out above)
    + payload.rehab_cushion
    + max(0, -cash_out_routi_conservative) (cash_to_refi_table_conservative)

cash_needed_floor =
    payload.earnest_money_deposit
    + cash_to_close_buy
    + payload.rehab_cushion
    + payload.monthly_utilities_until_rented
    + hml_amount * payload.HML_interest_rate * max(0, 30 - hml_interest_days_prepaid_at_purchase_closing) / 360 / 100
    + (payload.annual_property_taxes + payload.annual_insurance + payload.monthly_hoa * 12) * 30 / 360

total_cash_needed = cash_needed_through_refi + max(0, cash_needed_floor - cash_needed_through_refi)
                  = max(cash_needed_floor, cash_needed_through_refi)
```

Cash to Close (Buy)

closing_costs_buy.py

```
cash_to_close_buy =
    down_payment_cash                (= payload.down_payment / 100 * purchase_price)
    + closing_costs_buy_total
    + hml_and_holding.hml_points      (= payload.HML_points / 100 * hml_amount)
    + hml_and_holding.prepaid_interest_buy (= hml_amount * payload.HML_interest_rate * hml_interest_days_prepaid_at_purchase_closing / 360 / 100)
    - seller_tax_credit
    - payload.earnest_money_deposit

closing_costs_buy_total =
    payload.loan_charges_buy
    + (payload.recording_transfer_buy or 250 + 0.0055 * hml_amount + deed_transfer_tax_buy)
    + (payload.title_escrow_buy or 1000 standard / 2050, 2200, 2400 price tiers when we_pay_all)
    + (payload.online_notary_fee_buy or 250, if payload.online_notary_buy, else 0)
    + payload.other_closing_costs_buy

deed_transfer_tax_buy = 0.0070 * purchase_price if payload.title_mode_buy == "we_pay_all", else 0
hml_amount = purchase_price * (1 - payload.down_payment / 100) + construction_budget

seller_tax_credit = 0
                  with no buy_closing_date
                  = -payload.annual_property_taxes * (days_in_year - day_of_year + 1) / days_in_year if seller already paid (default: December closing)
                  = payload.annual_property_taxes * (day_of_year - 1) / days_in_year otherwise
```

Cash-Out Routi (Lowest ARV)

cash_out.py

```
cash_out_routi_conservative =  
    conservative_refi_loan_amount          (= lowest_arv * ltv;  lowest_arv = payload.lowest_arv_in_thousands * 1000, or 0.90 * arv)  
    - hml_payoff  
    - closing_costs_refi_total (at conservative_refi_loan_amount)  
    - prepaid_interest_refi (at conservative_refi_loan_amount)  
    - reserves_total
```

Stolen Money

rehab_draw.py

```
stolen_money = construction_budget - rehab_cost  
  
construction_budget = payload.construction_loan_budget_in_thousands * 1000  
rehab_cost = payload.rehab_cost_in_thousands * 1000 * (1 + payload.rehab_contingency_percent / 100)
```

Shared holding-period terms referenced above

hml_and_holding_costs.py

```
hml_interest = hml_amount * payload.HML_interest_rate * payload.days_until_refi / 360 / 100  
hml_interest_accrued_into_refi_payoff = hml_amount * payload.HML_interest_rate * hml_interest_days_accrued_into_refi_payoff / 360 / 100  
hml_interest_paid_monthly = hml_interest - prepaid_interest_buy - hml_interest_accrued_into_refi_payoff  
holding_costs = (payload.annual_property_taxes + payload.annual_insurance + payload.monthly_hoa * 12) * payload.days_until_refi / 360  
utilities_until_rented = payload.monthly_utilities_until_rented * payload.days_until_rented / 30  
pre_refi_rental_income = payload.rent * max(0, payload.days_until_refi - payload.days_until_rented) / 30
```